PH 101

Lecture 14

OUTLINE

- Hamiltonian formulation: worked out examples
- Phase Space
- Conclusion: Advantages of Lagrangian formalism over that of Newtonian formalism
- Mid-Semester Examination: Some points to remember!

LAST CLASS:

The Hamiltonian:

$$\boxed{H(\boldsymbol{q}, \boldsymbol{p}, t) = \dot{\boldsymbol{q}} \cdot \boldsymbol{p} - L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)}$$

Hamilton's equations of motion

$$\dot{q}_j = \frac{\partial H}{\partial p_j}, \qquad \dot{p}_j = -\frac{\partial H}{\partial q_j} \qquad (1 \leq j \leq n).$$

Worked out Examples

Example 1: Find the Hamiltonian and Hamilton's equations for the simple pendulum.

Solution

The **Lagrangian** for the simple pendulum is

$$L = \tfrac{1}{2}ma^2\dot{\theta}^2 + mga\cos\theta,$$

where θ is the angle between the string and the downward vertical, m is the mass of the bob, and a is the string length. The momentum p_θ conjugate to the coordinate θ is given by

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = ma^2\dot{\theta}$$

and this formula is easily inverted to give

$$\dot{\theta} = \frac{p_\theta}{ma^2}. \qquad (1)$$

The Hamiltonian H is then given by

$$H = \dot{\theta}\, p_\theta - L,$$

where $\dot{\theta}$ is given by equation (1) This gives

$$\begin{aligned} H &= \left(\frac{p_\theta}{ma^2}\right) p_\theta - \tfrac{1}{2} ma^2 \left(\frac{p_\theta}{ma^2}\right)^2 - mga\cos\theta \\ &= \frac{p_\theta^2}{2ma^2} - mga\cos\theta. \end{aligned}$$

This is the **Hamiltonian** for the simple pendulum. From H we can find Hamilton's equations. They are

$$\begin{aligned} \dot{\theta} &= \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{ma^2}, \\ \dot{p}_\theta &= -\frac{\partial H}{\partial \theta} = -mga\sin\theta. \end{aligned}$$

These are **Hamilton's equations** for the simple pendulum.

Example 2:

Spherical pendulum

The spherical pendulum is a particle of mass m attached to a fixed point by a light inextensible string of length a and moving under uniform gravity. It differs from the simple pendulum in that the motion is not restricted to lie in a vertical plane. Show that the Lagrangian is

$$L = \tfrac{1}{2}ma^2\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\phi}^2\right) + mga\cos\theta,$$

where the polar angles θ, ϕ are shown in Figure 1

Find the Hamiltonian and obtain Hamilton's equations. Identify any cyclic co-ordinates.

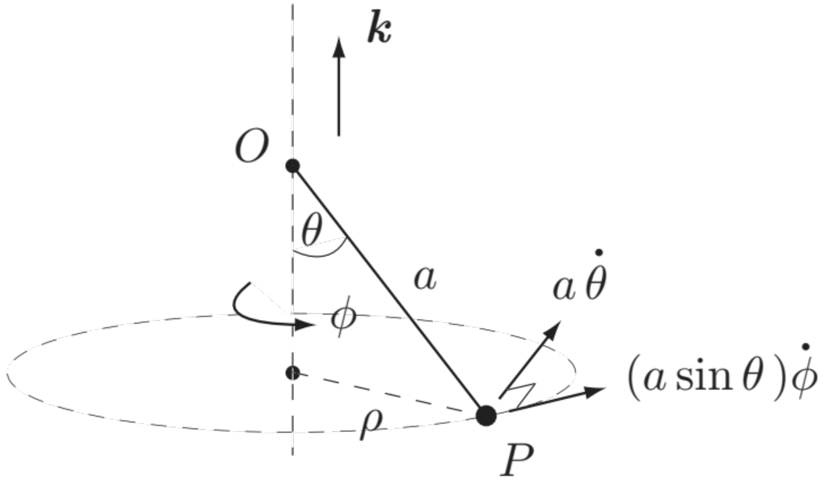


Fig. 1

Solution

In terms of the polar angles θ, ϕ, the system has **kinetic energy**

$$T = \tfrac{1}{2}m\left((a\dot{\theta})^2 + (a\sin\theta\,\dot{\phi})^2\right)$$

and **potential energy** $V = -mga\cos\theta$. Hence the **Lagrangian** of the system is

$$L(\theta,\phi,\dot{\theta},\dot{\phi}) = T - V = \tfrac{1}{2}ma^2\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\phi}^2\right) + mga\cos\theta.$$

The **conjugate momenta** p_θ, p_ϕ are then given by

$$p_\theta = \frac{\partial L}{\partial\dot{\theta}} = ma^2\dot{\theta},$$
$$p_\phi = \frac{\partial L}{\partial\dot{\phi}} = ma^2\sin^2\theta\,\dot{\phi}.$$

Since this system is conservative, the **Hamiltonian** is given by

$$
\begin{aligned}
H(\theta, \phi, p_\theta, p_\phi) &= T + V \\
&= \tfrac{1}{2} m a^2 \left(\left(\frac{p_\theta}{m a^2} \right)^2 + \sin^2\theta \left(\frac{p_\phi}{m a^2 \sin^2\theta} \right)^2 \right) - m g a \cos\theta \\
&= \frac{1}{2 m a^2} \left(p_\theta^2 + \frac{p_\phi^2}{\sin^2\theta} \right) - m g a \cos\theta.
\end{aligned}
$$

Hamilton's equations for the system are then given by

$$
\begin{aligned}
\dot{\theta} &= \frac{\partial H}{\partial p_\theta}, \qquad & \dot{p}_\theta &= -\frac{\partial H}{\partial \theta}, \\
\dot{\phi} &= \frac{\partial H}{\partial p_\phi}, \qquad & \dot{p}_\phi &= -\frac{\partial H}{\partial \phi},
\end{aligned}
$$

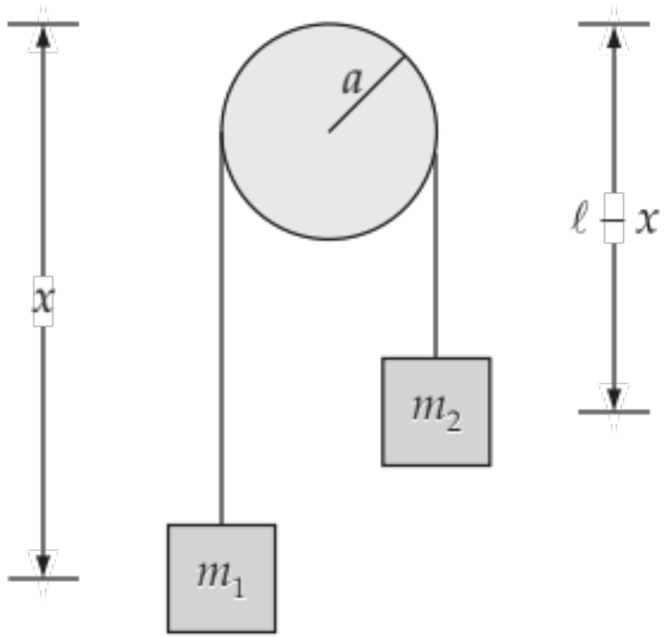
$$\dot{\theta} = \frac{p_\theta}{ma^2}, \qquad \dot{p}_\theta = \frac{p_\phi^2 \cos\theta}{ma^2 \sin^3\theta} - mga\cos\theta,$$

$$\dot{\phi} = \frac{p_\phi}{ma^2 \sin^2\theta}, \qquad \dot{p}_\phi = 0.$$

The coordinate ϕ does not appear in H and is therefore **cyclic**. As a result, the conjugate momentum p_ϕ is conserved. ■

Example 3:

ATWOOD MACHINE



The pulley can rotate and has moment of inertia I

Kinetic energy:

$$T = \frac{1}{2} m_1 \dot{x}^2 + \frac{1}{2} m_2 \dot{x}^2 + \frac{1}{2} I \frac{\dot{x}^2}{a^2}$$

Potential energy:

$$U = -m_1 g x - m_2 g(\ell - x)$$

For a change, let us denote potential energy by ***U*** *instead of the usual* ***V*** *!*

Now,

$$p_x = \frac{\partial L}{\partial \dot{x}} = \frac{\partial T}{\partial \dot{x}} = \left[m_1 + m_2 + \frac{I}{a^2} \right] \dot{x}$$

So,

$$\dot{x} = \frac{p_x}{\left[m_1 + m_2 + \frac{I}{a^2} \right]}$$

$$H = T + U$$

$$\boxed{H = \frac{p_x^2}{2\left[m_1 + m_2 + \frac{I}{a^2}\right]} - m_1 g x - m_2 g(\ell - x)}$$

The equations of motion are:

$$\boxed{\begin{aligned} \dot{x} &= \frac{\partial H}{\partial p_x} = \frac{p_x}{2\left[m_1 + m_2 + \frac{I}{a^2}\right]} \\ \dot{p}_x &= -\frac{\partial H}{\partial p_x} = m_1 g - m_2 g = g(m_1 - m_2) \end{aligned}}$$

HAMILTONIAN PHASE SPACE ((q, p)–space)

Suppose the mechanical system $\mathcal{S}$ has generalised coordinates $\boldsymbol{q}$, conjugate momenta $\boldsymbol{p}$, and Hamiltonian $H(\boldsymbol{q}, \boldsymbol{p}, t)$. If the initial values of $\boldsymbol{q}$ and $\boldsymbol{p}$ are known, then the subsequent motion of $\mathcal{S}$, described by the functions $\{\boldsymbol{q}(t), \boldsymbol{p}(t)\}$, is uniquely determined by Hamilton's equations. This motion can be represented geometrically by the motion of a 'point' (called a **phase point**) in Hamiltonian **phase space**. Hamiltonian phase space is a real space of $2n$ dimensions in which a 'point' is a set of values $(q_1, q_2, \ldots, q_n, p_1, p_2, \ldots, p_n)$ of the independent variables $\{\boldsymbol{q}, \boldsymbol{p}\}$. (Note that a point in Hamiltonian phase space represents not only the configuration of the system $\mathcal{S}$ but also its instantaneous momenta. This is the distinction between Hamiltonian phase space, which has $2n$ dimensions, and Lagrangian configuration space, which has n dimensions.) Each motion of the system $\mathcal{S}$ then corresponds to the motion of a phase point through the phase space.

Example 4:

Suppose that $\mathcal{S}$ has the single coordinate q and Lagrangian

$$L = \frac{\dot{q}^2}{4} - \frac{q^2}{9}.$$

Find the paths in Hamiltonian phase space that correspond to the motions of $\mathcal{S}$.

Solution

The conjugate momentum $p = \partial L/\partial \dot{q} = \frac{1}{2}\dot{q}$, and the Hamiltonian is

$$H = \dot{q}\,p - L = (2p)\,p - \frac{1}{4}(2p)^2 + \frac{q^2}{9} = p^2 + \frac{q^2}{9}.$$

Hamilton's equations for $\mathcal{S}$ are therefore

$$\dot{q} = 2p, \qquad \dot{p} = -2q/9.$$

On eliminating p, we find that q satisfies the SHM equation

$$\ddot{q} + (4q/9) = 0.$$

The general solution of the Hamilton equations for $\mathcal{S}$ is therefore

$$q = 3A\cos((2t/3) + \alpha), \qquad p = -A\sin((2t/3) + \alpha),$$

where A and α are arbitrary constants. These are the parametric equations of the **paths** in phase space, the parameter being the time t; each path corresponds to a possible motion of the system $\mathcal{S}$. Some typical paths are shown in Figure 1 For this system , every motion is periodic so that the paths are *closed* curves in the (q, p)-plane. (They are actually concentric similar ellipes.) The arrows show the direction that the phase point moves along each path as t increases. ■

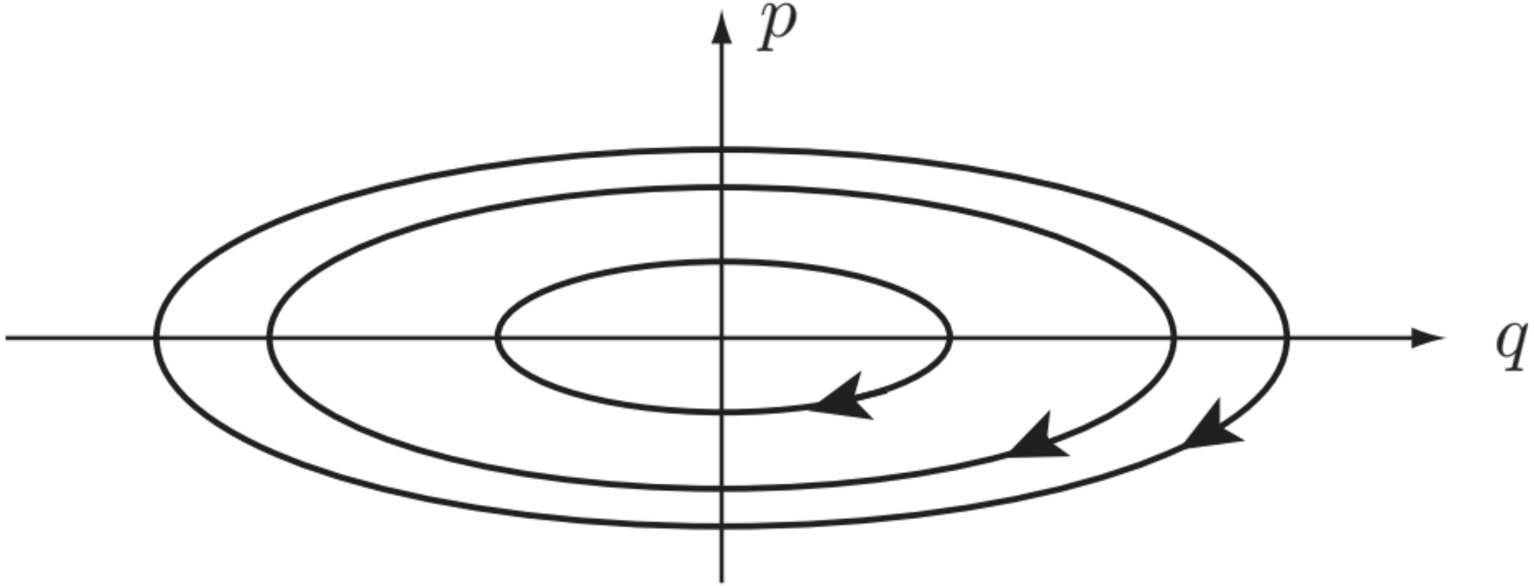


FIGURE 1 Typical paths in the phase space (q, p) corresponding to motions of a system $\mathcal{S}$ with Hamiltonian $H = p^2 + q^2/9$. The arrows show the direction that the phase point moves along each path as t increases.

FINAL WORDS-I

ADVANTAGES OF LAGRANGIAN FORMALISM OVER NEWTONIAN MECHANICS

In the Newtonian mechanics, the equations of motion involve vector quantities like force, momentum etc. which increase complexity in solving the problems. This approach also cannot avoid constraints present in a problem. These forces of constraints, if not known, make the solution of the problem difficult and even if they are known, the use of rectangular or other commonly used coordinates may make the solution of the problem to be impossible. These drawbacks are removed in the Lagrangian mechanics, where the technique involves scalars, like potential and kinetic energies, instead of vectors. The use of generalized coordinates in the Lagrangian formulation often allows automatically for the constraints. In this formulation, the difficulty in solving the problems is many times much reduced, when any quantity like momentum or $(\text{length})^2$ is taken as a generalized coordinate instead of rectangular or commonly used coordinates. Further the form of the Lagrange's equations of motion remains invariant under any generalized coordinate transformation.

FINAL WORDS-II

ADVANTAGES OF HAMILTONIAN FORMALISM:

Transition from Classical Mechanics to Quantum Mechanics

In classical physics, there are physically measurable quantiles like position, momentum, energy and so on. In Quantum Mechanics (QM), corresponding to every physically measurable quantity in CM, there is an operator. For example, corresponding to linear momentum $\vec{p}$, we have the linear momentum operator $\hat{p}$. In QM, the state of a system is given by the so-called wave function or state-vector ψ. If you want to know about the linear momentum of the quantum system, you need to operate on the state-vector ψ by the linear momentum operator $\hat{p}$:

$$\hat{p}\psi = p\psi,$$ where p is real valued and is the linear momentum.

Similarly if you want to now about the energy of the quantum system, you need to operate on the state-vector ψ by the Hamiltonian operator $\widehat{H}$:

$$\widehat{H}\psi = E\psi,$$ where E is real valued and is the Energy eigenvalue of the system!

Knowing the Hamiltonian H of the system is one of the most crucial steps, and you can go over to QM easily by treating it as an operator, $\widehat{H}$!

Moreover, don't forget the key equation of QM:

$$\boxed{i\hbar\frac{\partial\psi}{\partial t} = \widehat{H}\psi}$$

The Schrodinger Equation

See the Hamiltonian there?

Mid-Semester Examination: Some points to remember!

- The full syllabus, whatever is taught from Lecture 1-Lecture 14, will be covered.
- No derivation will be asked, e.g. you will not be asked to derive Euler-Lagrange's equation!
- You will be given a Question cum Answer Booklet, where you have to write the final answer.
- Detailed calculations can be done in the rough sheet to be provided to you.
- You have to return both the 'Question cum Answer Booklet' and the 'rough-sheet' to the invigilator.
- You are not allowed to carry Mobile Phone to the examination Hall. A direct 'F' grade if found otherwise.
- Any kind of cheating will land you up in lots of trouble, including termination from the institute!

A Sample Question and expected Answer in the **Question cum Answer Booklet** !

Q. 1. Consider a simple pendulum consisting of mass m attached to a string of length l.

(a) Write the Lagrangian for the system.

(b) Find the generalized momenta.

(c) Find the Hamiltonian.

(d) Write down the Hamilton's equations of motion.

Answer:

(a)

$$L = \frac{1}{2} m\ell^2 \dot{\theta}^2 + mg\ell \cos\theta$$

(b)

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m\ell^2 \dot{\theta}$$

(c)

$$H = T + U = \frac{p_\theta^2}{2m\ell^2} - mg\ell \cos\theta$$

(d)

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{m\ell^2} \text{ and } \dot{p}_\theta = -\frac{\partial H}{\partial \theta} = -mg\ell \sin\theta$$

Important Formulae/Info.

- Euler-Lagrange's eq^n:

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0$$

- (necessary condition for $y(x)$ to make $S = \int f(y, y', x)\,dx$ stationary)

- Lagrange's equation of motion:

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = 0$$

- Energy function:

$$E = \sum_i \dot{q}_i p_i - L$$

- Hamiltonian:

$$H(q_i, p_i, t) = \sum_i \dot{q}_i p_i - L$$

- Hamilton's equations of motion:

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \; ; \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Wish You All the Best in Life !